

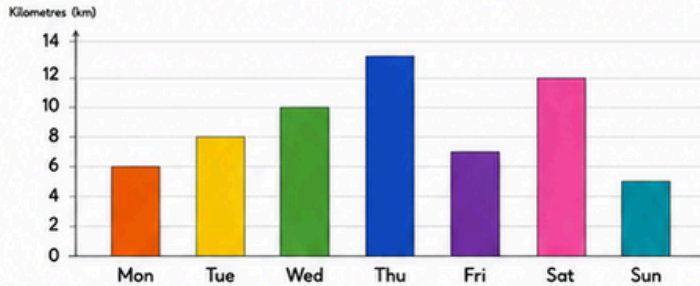
Interpret Richer Graphs and Summary Measures

SDA UE Step 7

★ I can interpret graphs and use summary measures to answer questions.

1 Let's Look A school recorded the number of kilometres students walked each day for a week.

Kilometres Walked Each Day



Summary Measures

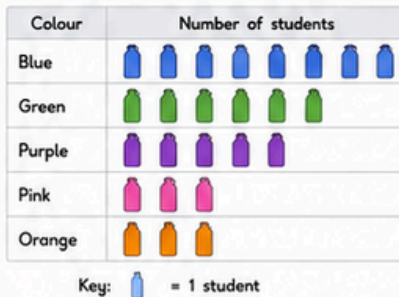
Maximum	13 km
Minimum	5 km
Range	8 km
Total	61 km
Mean (average)	8.7 km (about 9 km)

Talk About It

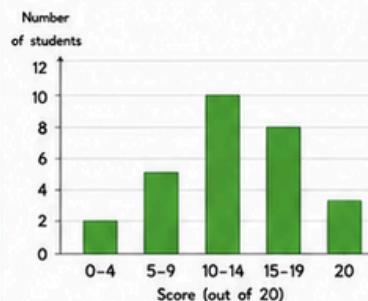
- On which day was the greatest distance walked?
- On which day was the smallest distance walked?
- What is the range of the data?
- What is the mean (average) kilometres walked?
- How many more kilometres were walked on Thursday than Sunday?

2 Let's Try Interpret the graphs and answer the questions.

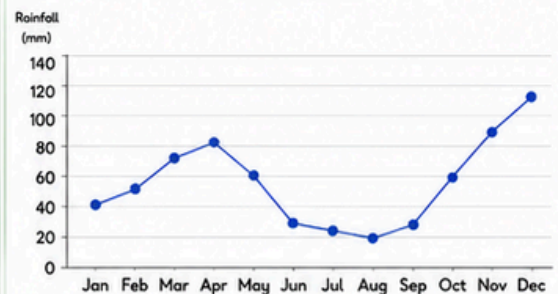
A. Favourite Water Bottle Colour



B. Scores in a Maths Quiz



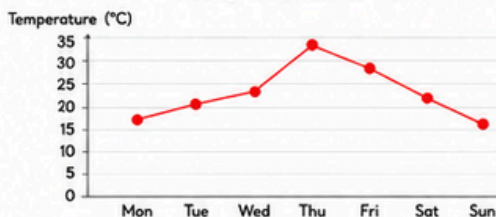
C. Monthly Rainfall



- | | | |
|--|--|---|
| <p>a. Which colour is the most popular? _____</p> <p>b. Which colour is the least popular? _____</p> <p>c. How many students chose purple or pink? _____</p> <p>d. How many students were surveyed in total? _____</p> | <p>a. Which score group has the most students? _____</p> <p>b. Which score group has the least students? _____</p> <p>c. How many students scored 10 or more? _____</p> <p>d. How many students sat the quiz in total? _____</p> | <p>a. In which month was the most rainfall? _____</p> <p>b. In which month was the least rainfall? _____</p> <p>c. What is the range of the monthly rainfall? _____</p> <p>d. What is the mean (average) rainfall for the year? _____ mm (about _____ mm)</p> |
|--|--|---|

3 Let's Do Answer the questions using the graph and summary measures.

Weekly Temperatures (°C)



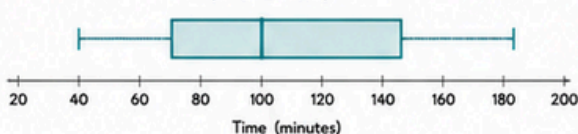
Summary Measures

Maximum	31°C
Minimum	17°C
Range	14°C
Mean (average)	24°C

- a. What was the temperature on Thursday? _____ °C
- b. On which day was the temperature 22°C? _____
- c. On which day was the greatest temperature recorded? _____
- d. On which day was the smallest temperature recorded? _____
- e. What is the range of the temperatures? _____ °C
- f. What is the mean (average) temperature? _____ °C
- g. On how many days was the temperature 20°C or higher? _____
- h. Write one sentence describing the trend in temperature. _____

4 Challenge Use the box-and-whisker plot to answer the questions.

Time (minutes) taken to read a book by 20 students
(lower = faster)



- a. What is the minimum time? _____ minutes
- b. What is the maximum time? _____ minutes
- c. What is the median time? _____ minutes
- d. What is the lower quartile (Q1)? _____ minutes
- e. What is the upper quartile (Q3)? _____ minutes
- f. What is the interquartile range (IQR)? _____ minutes

Think About It

Why is the median a useful measure when comparing times in this data set?

My Learning Check

How did I go today?



I can do it!



Almost there!



I need more help.



This was hard.

Circle a face.

Parent Observation (optional)

Tick what you saw today.

- ☐ Interpreted bar, pictograph, line and box plots
- ☐ Used summary measures (mean, range, min, max, median)
- ☐ Compared categories and described trends
- ☐ Answered questions using data
- ☐ Needed support

Notes: _____